

# Homework 4

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## Exercise 1

Let  $A$  be a  $2 \times 2$  upper triangular matrix. It then has the form:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{bmatrix}$$

Its characteristic polynomial is then:

$$\det(A - \lambda I) = (a_{11} - \lambda)(a_{22} - \lambda)$$

For this to be zero,  $\lambda = a_{11}, a_{22}$ , so the eigenvalues are exactly the diagonal elements of  $A$ . Now suppose that for all  $n \times n$  upper triangular matrices, the eigenvalues are exactly the diagonal elements of  $A$ . We now consider a  $(n+1) \times (n+1)$  upper triangular matrix. It then has the form:

$$A = \begin{bmatrix} a_{11} & B \\ 0 & A' \end{bmatrix}$$

where  $B$  is a  $1 \times n$  matrix and  $A'$  is a  $n \times n$  upper triangular matrix. Its characteristic polynomial is then:

$$\det(A - \lambda I) = (a_{11} - \lambda) \det(A' - \lambda I)$$

For this to be zero, either  $\lambda = a_{11}$  or  $\lambda$  is a diagonal element of  $A'$ . Note that the diagonal elements of  $A'$  are all but the first diagonal element of  $A$ , so the eigenvalues are exactly the diagonal elements of  $A$ . By induction, the eigenvalues of all upper triangular matrices are exactly its diagonal elements.

## Exercise 2

### Part (a)

Let  $A, B$  be similar  $n \times n$  matrices. Then  $B = P^{-1}AP$  for some  $n \times n$  matrix  $P$ , or equivalently,  $A = PBP^{-1}$ . Suppose  $\lambda$  is an eigenvalue of  $A$ ; then  $Av = \lambda v$ , where  $v$  is the corresponding eigenvector. Then:

$$\begin{aligned}Av &= (PBP^{-1})v = \lambda v \\BP^{-1}v &= \lambda P^{-1}v \\B(P^{-1}v) &= \lambda(P^{-1}v)\end{aligned}$$

Consequently,  $\lambda$  is also an eigenvalue of  $B$ . By symmetry, the eigenvalues of  $A$  are exactly the eigenvalues of  $B$ .

### Part (b)

If  $v_A$  is an eigenvector of  $A$  and  $v_B$  is an eigenvector of  $B$ , then  $v_B = P^{-1}v_A$ .

## Exercise 3

### Part (a)

Let  $u, v \in \mathbb{R}^n$  with  $u^T v = 1$ . Suppose  $A = uv^T$ ; note that  $A$  is an outer product and has  $\text{rank}(A) = 1$ , so it only has one nonzero eigenvalue. Then:

$$\begin{aligned} Au &= uv^T u \\ &= u(1) = u \end{aligned}$$

The remaining nonzero eigenvalue is 1, with an eigenvector of  $u$ .

### Part (b)

By the power iteration method:

$$x^{(k)} = \frac{A^k x^{(0)}}{\|A^k x^{(0)}\|}$$

Note that:

$$\begin{aligned} A^2 &= (uv^T)^2 \\ &= uv^T uv^T = u(1)v^T \\ &= uv^T = A \end{aligned}$$

By induction,  $A^k = A$  for all  $k$ . Now, since  $A$  has only one nonzero eigenvalue,  $x^{(0)} = \alpha u + w$ , where  $w \in \text{null}(A)$  and for some constant  $\alpha$ . Then:

$$\begin{aligned} x^{(1)} &= Ax^{(0)} = A(\alpha u + w) \\ &= uv^T(\alpha u) + Aw \\ &= \alpha u \end{aligned}$$

By induction,  $x^{(k)} = x^{(0)}$  for all  $k$ , and power iteration converges in one iteration.

## Exercise 4

The matrix:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

has eigenvalues  $\lambda_1 = 1$  and  $\lambda_2 = -1$ , with eigenvectors  $v_1 = (1, 1)^T$  and  $v_2 = (1, -1)^T$ , respectively.  $x^{(0)}$  can be decomposed as  $x^{(0)} = \alpha_1 v_1 + \alpha_2 v_2$ . Then:

$$A^k x^{(0)} = \alpha_1 (1)^k v_1 + \alpha_2 (-1)^k v_2$$

The result of power iteration then oscillates between  $\alpha_1 v_1 + \alpha_2 v_2$  and  $\alpha_1 v_1 - \alpha_2 v_2$ , and does not converge. This occurs because the two eigenvalues have equal magnitude.

## Exercise 5

Generalizing the QR algorithm to find eigenvectors and eigenvalues involves a shift in each iteration:

$$\begin{aligned} A_0 &= A \\ A_k - \sigma_k I &= Q_k R_k \\ A_{k+1} &= R_k Q_k + \sigma_k I \end{aligned}$$

Let  $A$  be a  $n \times n$  matrix, with  $A = Q_0 R_0$ . Then:

$$\begin{aligned} A_1 &= R_0 Q_0 + \sigma_0 I \\ &= Q_0^T Q_0 R_0 Q_0 + \sigma_0 Q_0^T Q_0 \\ &= Q_0^T (Q_0 R_0 + \sigma_0 I) Q_0 \\ &= Q_0^T (A_0 - \sigma_0 I + \sigma_0 I) Q_0 = Q_0^T A_0 Q_0 \end{aligned}$$

Therefore  $A_1$  is similar to  $A$ , with  $A_1 = P^{-1} A P = P^T A P$  for some orthogonal  $P$ . Suppose  $A_k$  is similar to  $A$  for all  $k$ , with  $A_k = P_k^T A P_k$  for some orthogonal  $P_k$ . Then:

$$\begin{aligned} A_{k+1} &= R_k Q_k + \sigma_k I \\ &= Q_k^T Q_k R_k Q_k + \sigma_k Q_k^T Q_k \\ &= Q_k^T (Q_k R_k + \sigma_k I) Q_k \\ &= Q_k^T (A_k - \sigma_k I + \sigma_k I) Q_k = Q_k^T A_k Q_k \\ &= Q_k^T (P_k^T A P_k) Q_k \\ &= (P_k Q_k)^T A (P_k Q_k) \end{aligned}$$

Note that  $P_k Q_k$ , being the product of two orthogonal matrices, is orthogonal. Therefore  $A_k$  is similar to  $A$ , with  $A_1 = P'^{-1} A P' = P'^T A P'$  for some orthogonal  $P'$ . By induction,  $A_k$  is similar to  $A$  for all  $k$ .

## Exercise 6

### Part (a)

```
1 n = 10;
2 k_max = 100;
3 A = 2 * eye(n) - diag(ones(n - 1, 1), 1) - diag(ones(n - 1, 1), -1);
4
5 % power iteration
6 x = [1; zeros(n - 1, 1)];
7 for k = 1:k_max
8     x = A * x;
9     lambda = norm(x);
10    x = x / lambda;
11 end
```

The computed eigenvalue is  $\lambda = 3.9190$  and the corresponding eigenvector is:

$$v = \begin{bmatrix} 0.1210 \\ -0.2320 \\ 0.3239 \\ -0.3891 \\ 0.4225 \\ -0.4216 \\ 0.3866 \\ -0.3206 \\ 0.2291 \\ -0.1193 \end{bmatrix}$$

### Part (b)

```
1 n = 10;
2 k_max = 100;
3 A = 2 * eye(n) - diag(ones(n - 1, 1), 1) - diag(ones(n - 1, 1), -1);
4
5 % QR, no shift
6 A_k = A;
7 off_diag_norm = zeros(k_max, 1);
8
9 for k = 1:k_max
10    [Q, R] = qr(A_k);
11    A_k = R * Q;
12    off_diag_norm(k) = norm(A_k - diag(diag(A_k)));
13 end
```

The computed eigenvalue is  $\lambda = 3.9190$ . The off-diagonal elements decay gradually.

### Part (c)

```
1 n = 10;
2 k_max = 100;
```

```

3 | A = 2 * eye(n) - diag(ones(n - 1, 1), 1) - diag(ones(n - 1, 1), -1);
4 |
5 | % QR, shift
6 | A_k = A;
7 | off_diag_norm = zeros(k_max, 1);
8 |
9 | for k = 1:k_max
10 |     sigma = A_k(n, n);
11 |     [Q, R] = qr(A_k - sigma * eye(n));
12 |     A_k = R * Q + sigma * eye(n);
13 |     off_diag_norm(k) = norm(A_k - diag(diag(A_k)));
14 | end

```

The computed eigenvalue is  $\lambda = 2.0000$ . The off-diagonal elements do not decay, likely because the diagonal elements of  $A$  are identical.